T-World with intracellular ATP/ADP dependence: summary and lower-bound effect on the Ca^{2+} transient

Branch `LowAtp`. Full write-up, figures and references: README.md (branch `LowAtp`).

What was added to the model

All changes are opt-in. With the defaults, the model output is **bitwise identical** to the current T-World for the endo, epi and mid cell types (checked with `isequal` on the full trajectories).

- **New parameters**
  - `ATP_i` and free `ADP_i` (mM), clamped during a run.
  - The reference state `ATP_ref` = 8 mM and `ADP_ref` = 93.7 µM. `ADP_ref` is the human creatine kinase equilibrium (Weiss et al. 2005 PCr/ATP/total-Cr pools, K_CK = $1.66 \cdot 10^9$ M^{-1}, pH 7.05), computed by the new `getCKEquilibriumADP.m`.
  - `IKATP_Multiplier`, `KATP_Km_Multiplier` and `ca50_Multiplier`, all 1 by default.
- **Pump scaling (SERCA V_max, I_NaK, I_pCa).** The Cortassa et al. (2006) ATP/ADP factors, checked against the appendix (Eqs. A35, A46, A50).
  - Each factor is **divided by its value at the reference state**, so it is exactly 1 there.
  - The I_pCa factor is a sum with maximum 2, so normalising it is essential.
- **I_KATP.** The human version of Ferrero et al. (1996) by Mena Tobar, Ferrero & Rodriguez Matas (CinC 2015): g_0 = 4.5, K_m = α(35.8 + 17.9·ADP^0.256) µM, H = 1.3 + 4.44·e^(−0.09·ADP), α = 0.32/0.86/1.0 for endo/mid/epi.
  - Only the open fraction **in excess of the reference** is added. The resting K_ATP conductance (~0.07 % open in endo) is taken to be absorbed in the calibrated background K^+ currents.
  - I_KATP enters I_K_tot, so it acts on both V_m and $[K^+]_i$.
- **`ca50_Multiplier`** scales the Land model TnC ca50, to probe myofilament Ca^{2+} sensitivity.
- **New outputs.** `getCurrentsStructure` gains `IKATP` and the four factors `fATP_*` (output indices 36–40).
  - `model_TWorld` takes a trailing `atpParams` argument, which defaults when it is omitted.
  - Existing callers of the model are unaffected.

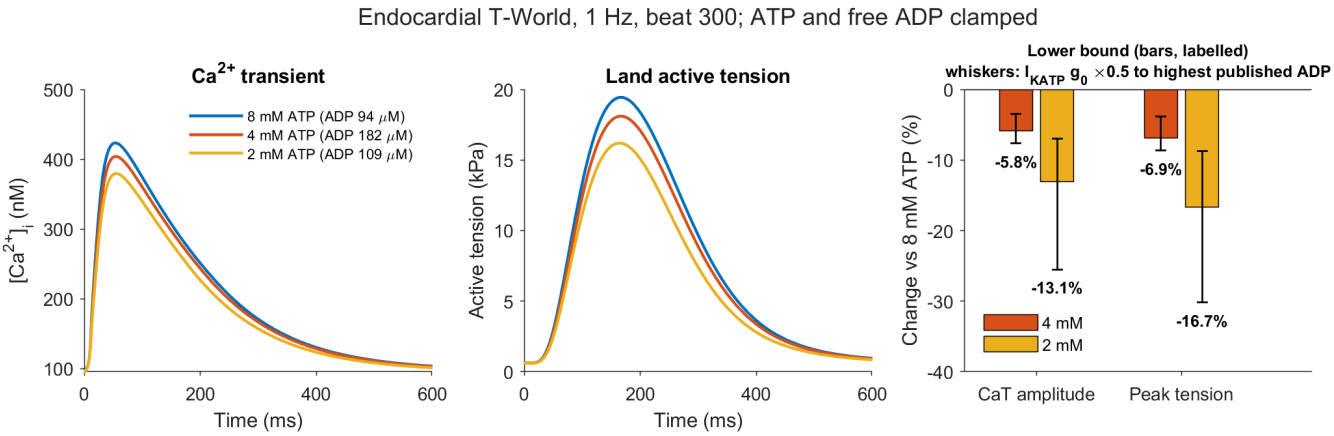
Question and approach

What is the **smallest** change in the Ca^{2+} transient (CaT) when ATP falls from 8 mM to 4 or 2 mM, given current data? Every uncertain choice was set in the effect-reducing direction:

- **Free ADP.** The lowest published ischaemic free ADP at that fraction of ATP: 182 µM at 50 % and 109 µM at 25 % (guinea pig; Ferrero et al. 2023 / Terkildsen et al. 2007).
  - Rat data (Bak & Ingwall 1994): 284 µM and 767 µM.
  - CK equilibrium cannot predict these values, because PCr is exhausted before ATP halves.
- **ADP_ref.** The human value of 94 µM, which is high compared with 15–39 µM in animals.
- **Cell type.** Endocardial, the least K_ATP-sensitive (α = 0.32).
- **ADP in the pump factors.** Free ADP, although Cortassa's constants use total ADP.
- **Omitted mechanisms.** Mechanisms that would only increase the effect are omitted:
  - Pi in SERCA thermodynamics;
  - MgADP myofilament sensitisation;
  - ATP activation of RyR2.

Protocol: endo cell at 1 Hz, nucleotides clamped, last of 300 beats (150 beats for sweeps).

Results



Summary figure. Ca^{2+} transient (left) and Land model active tension (middle) at 8, 4 and 2 mM ATP, in the lower-bound scenarios (lowest published ischaemic ADP). Right: change in CaT amplitude and peak tension relative to 8 mM. Bars are the lower bound. Whiskers run from I_KATP g_0 ×0.5 (smallest effect) to the highest published ADP (largest). Tension falls more than the CaT, because the Land model's force–Ca^{2+} relation is steep: −5.8 % CaT gives −6.9 % tension, and −13.1 % gives −16.7 %. Figure script: `makeSummaryFigure.m`.

Changes are relative to 8 mM at matched myofilament sensitivity.

	4 mM ATP	**2 mM ATP**
CaT amplitude, lower bound	**−5.8 %**	**−13.1 %**
CaT amplitude, I_KATP g_0 halved	−3.5 %	−7.0 %
CaT amplitude, I_KATP blocked (pumps only)	−1.1 %	−0.3 %
CaT amplitude, highest published ADP	−7.6 %	−25.5 %
APD_{90} (lower bound)	−8.2 %	−21.0 %
Peak active tension (lower bound)	−6.9 %	−16.7 %
CaT decay τ (lower bound)	+0.4 %	−3.8 %
Excess K_ATP open fraction	0.13 %	0.36 %
f_SERCA / f_NaK / f_pCa	0.980 / 0.996 / 0.986	0.990 / 0.994 / 0.959

1. **The effect is almost entirely electrical.**
   - Opening a fraction of a percent of K_ATP channels shortens the AP. That lowers L-type Ca^{2+} entry and, slightly, SR load (−1 %).
   - The pumps remain >96 % saturated, so the direct effect of ATP on Ca^{2+} handling is ≈1 %.
2. **Decay kinetics are not a robust marker.** In the lower-bound scenarios, τ, $CaTD_{50}$ and $CaTD_{90}$ change by less than 4 %; even at the highest published ADP, $CaTD_{90}$ changes by at most +7 %.
3. **Myofilament Ca^{2+} sensitivity is independent of the ATP effect but can mask it.**
   - Across ca50 ×0.9–2, the relative ATP effect stays at −5.8 to −6.1 % (4 mM) and −13.1 to −13.7 % (2 mM).
   - The absolute CaT is strongly ca50-dependent through TnC buffering: ca50 ×1.25 gives +14 % amplitude and −13 % τ.
   - Desensitisation of about ×1.1 (4 mM) or ×1.27 (2 mM) is enough to cancel the absolute reduction. Acidosis desensitises more than this (Palmer & Kentish 1994). The observable CaT in acidotic ischaemia may therefore not fall, even though the ATP-attributable component does.

Limitations

- **The bound depends on K_ATP.**

    - Without functional K_ATP channels, the guaranteed effect is ≈1 %.
    - The human gating rests on recombinant channel data: Babenko et al. (1998), IC_{50} ≈ 20 µM and Hill ≈ 1. Our H = 1.3 is the conservative choice.
    - The parameters come from a 4-page proceedings paper, fitted in TP06. g_0 is printed as "mS"; mS/µF is inferred.
    - The Mg^{2+}/Na^{+} block and Q_{10} constants of Ferrero et al. (1996) are from secondary sources and are unverified.
    - The rise in free Mg^{2+} as MgATP falls is not modelled. It would strengthen the Mg^{2+} block, which is partly covered by the g_0 ×0.5 row.
    - Sanity check: Ferrero et al. (1996) report ~0.6 % opening → 50 % APD shortening. Here 0.36 % gives 21 %, so the current is not overstated.
- **ADP data.** All ischaemic ADP values are animal and CK-derived, and they are semi-quantitative at acidic pH. Cortassa's constants are in-vitro, guinea pig/rat.
- **Not modelled:**
    - Pi;
    - extracellular K^{+} accumulation;
    - direct acidosis effects on channels and exchangers;
    - L-type rundown;
    - time-varying ischaemia (states are clamped);
    - $[Mg^{2+}]_i$ changes.
- **$[Na^{+}]_i$ drift.** $[Na^{+}]_i$ still drifts by 0.1 mM at 300 beats; the CaT is stable after about 50 beats.
- **Falsifiable prediction.**
    - In human myocytes dialysed with 2 mM vs 8 mM MgATP (~0.1 mM ADP, pH and Mg^{2+} clamped), the CaT amplitude should fall by at least ~7–13 %.
    - Glibenclamide should abolish that fall.

Questions for you (T-World)

- Is normalising to the reference state the right way to keep the calibrated baseline? The alternative is re-fitting the maximal rates under explicit nucleotide factors.
- Should the TP06-derived K_ATP g_0 be re-scaled to T-World's repolarisation reserve? If so, is there human APD-under-pinacidil (or metabolic-inhibition) data you would calibrate against?